

BELT: Expert-in-the-Loop Validation of Sensor-Based Bruxism Detection

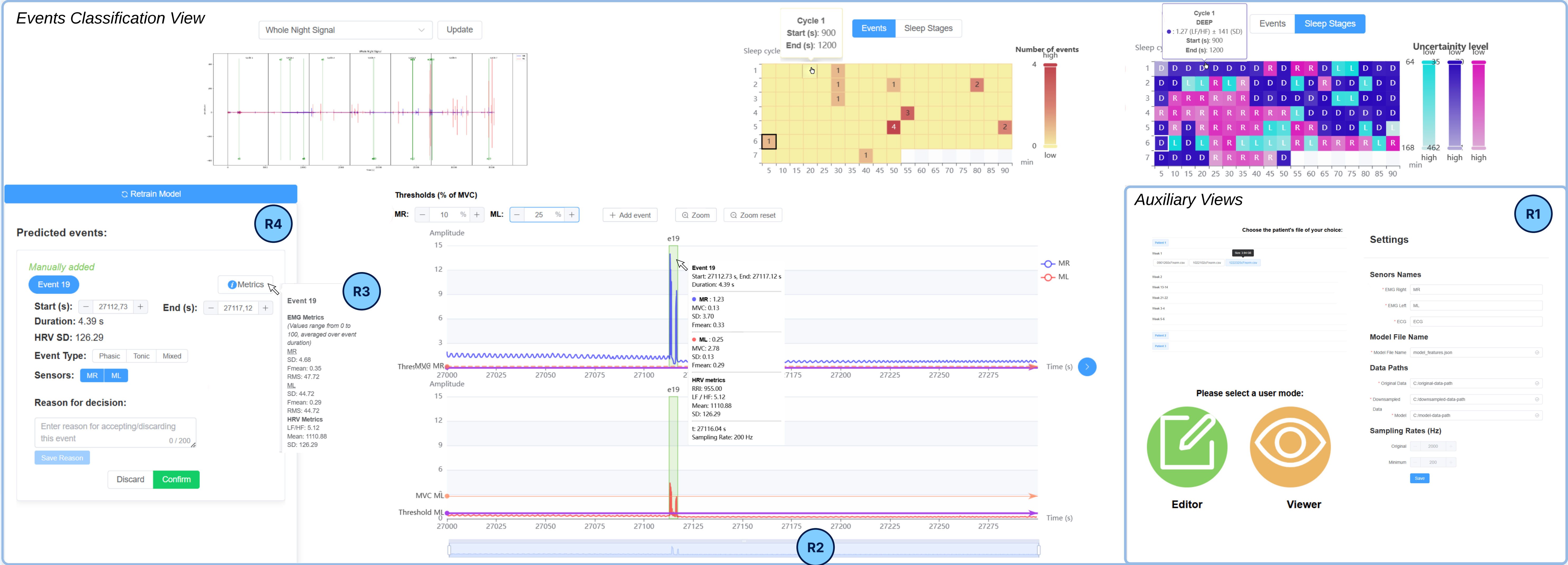


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PROBLEM: Bruxism events in at-home sleep studies *lack gold-standard labels*, and EMG signals are noisy and highly variable.

APPROACH: We use an EMG event prediction model to generate candidate events, which are then *reviewed and corrected* in a visual-interactive labeling interface built to support clinician sensemaking (XGBoost [1]).

CONTRIBUTION: *BELT captures expert-verified labels*, enabling model retraining and better *alignment* of predictions with clinical interpretation.



R1. Hierarchical information access Role-specific modes to access patient data, with hierarchical navigation of recordings to support exploration

R2. Time Resolution Control Multi-resolution pattern discovery enabled by static overviews paired with linked, zoomable short interval views, navigable by interactive heatmaps

R3. Feature-Level Contextualization Feature-level tooltips for both raw data and metrics, contextualizing predictions and supporting interpretability and comparability across recordings

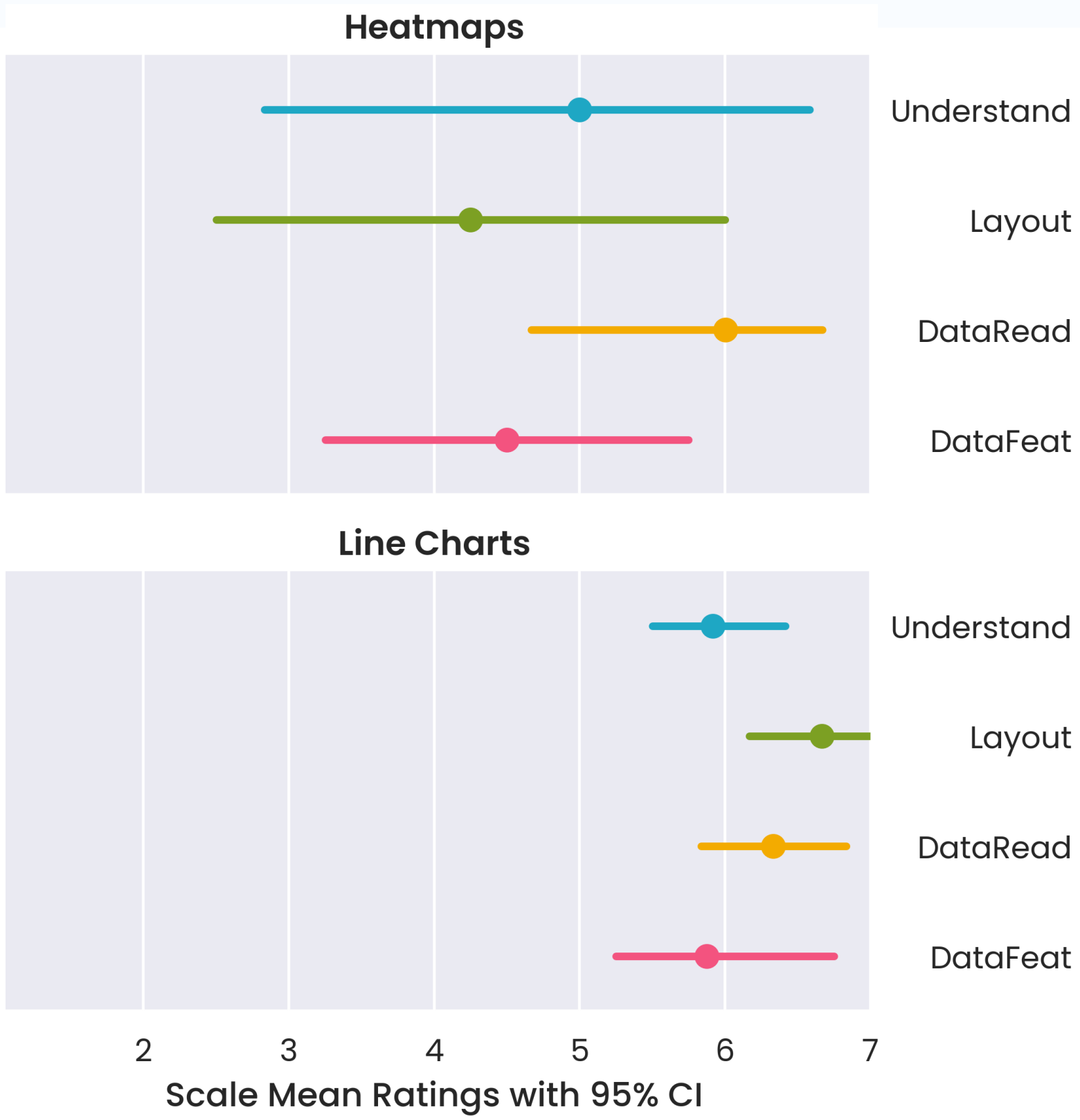
R4. Human-in-the-Loop Event Review and Labelling Expert-driven confirmation, editing, and annotation of predicted events, enabling transparent validation, retraining, and the capture of insights and model provenance

We piloted a study of BELT on 4 clinicians, and we observed...

High variance in results: substantial variance between participants evaluating visualization-assisted task performance (PREVis [2])

Mixed performance of the heatmap: despite initial unintuitive design, the heatmap was found to help participants identify patterns efficiently, and reduce cognitive overload during analysis

Takeaways: The question persists of how to effectively design interactive, visual navigation of long, high-frequency EMG recording data. Participant comments suggest that alternative or improved heatmap navigation may enhance usability and pattern recognition



Experts responded positively when system cues aligned with domain logic but grew skeptical in the absence of contextual explanations for their intuition...

Q1: How can we effectively support experts in making confident evaluation of modelling results under uncertainty?

Future work could explore domain-aligned, context-sensitive explanations that move beyond generic feature importance toward more intuitive insights...

Q2: How do we achieve the right balance between visual complexity and transparency in navigating high-frequency sensor data?

References

[1] T. Chen and C. Guestrin. Xgboost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, p. 785–794. ACM, Aug. 2016. doi:10.1145/2939672.2939785

[2] A.-F. Cabouat, T. Ho, P. Isenberger, and T. Isenberger. Previs: Perceived readability evaluation for visualizations. IEEE Transactions on Visualization and Computer Graphics, pp. 1–11, 2024. doi:10.1109/TVCG.2024.3456318